

**Statement of Work
BLM-Alaska
Contract No.**

**LIDAR Survey of Closed Mining Claims
Ruby Mining District**

1.0 Background

The Central Yukon Field Office of the Bureau of Land Management (BLM) has identified six areas of closed, excluded and unconveyed federal lands (CEUs) in the Ruby Mining District. CEUs are areas of federal land, typically former mining claims, surrounded by land managed by other entities, were selected by the State for conveyance but were not conveyed as part of the larger land conveyance processes due to encumbrances on the land at the time of conveyance. Encumbrances include active mine operations, unreclaimed closed mine operations, solid waste, and abandoned mine land (AML) features. This Statement of Work (SOW) has been created to assist in fulfilling the inventory requirements for the AML program. The data provided by the aerial survey will be interpreted by the BLM's Hazard Management and Resource Restoration (HMRR) specialist and the Cultural Resources Specialist/ Archaeologist who will look for structures, vehicles, and landscape contours to guide future on-ground inventory work. Areas with no features identified in the LIDAR data can be eliminated from on-ground inventory, reducing time and costs for that phase of work.

There are six groups of CEUs within the Ruby Mining District along the Ruby-Poorman Road. The CEUs are all closed and abandoned federal mining claims that have been partially inventoried over the last decade both on the ground and from aircraft. These inventories have been limited by time, weather, transportation, and conditions on the ground including vegetation overgrowth, flooding caused by beaver activity, and lack of trail maintenance due to disuse, among other issues. Historical documentation and aerial imagery of these areas is not extensive, and the overgrowth of vegetation has made walking through or seeing many areas almost impossible. Each consecutive inventory conducted has revealed additional features not observed in previous inventories. Though this method of partial inventories is revealing additional features, it is not a quick process and is often affected by lack of knowledge on where features may be. The LIDAR survey will allow the mapping of the entire area prior to arriving on the site and will allow for more informed and targeted field work. Several of the claims within these CEUs have been on the State Environmental Cleanup Action Plan (SECAP) as a priority for clean up in the last five years. The survey of these areas will move forward the cleanup process and allow for the lands to eventually be ready for conveyance to the State of Alaska.

2.0 Objective

The objective of this work is to generate a dataset that will assist in the completion of the inventory of abandoned mine land features and by extension, cultural resource features remaining on the six sites, as applicable. The data will be used by the HMRR/ AML and Cultural Resources programs to identify structures, vehicles, and contours in the landscape that help determine the historic use of the land. By identifying these features in the data provided by the LIDAR survey ahead of arriving on site, on-ground inventories in the near future can be targeted to these feature locations. The survey of these lands will allow the HMRR/ AML program to work jointly with the Cultural Resources/ Archaeology program to identify features that are hazardous to the environment and/or are not culturally significant and therefore should be removed. A removal planned by these programs will ensure that no historical or culturally significant features are affected and that all solid and hazardous wastes will be removed from the environment and the dataset from the LIDAR will assist removal contractors and ensure that historically significant cultural resources are avoided.

This action will assist in satisfying the inventory requirements for the Abandoned Mine Lands Program and the Cultural Resources Program. To achieve the goal of eliminating CEUs within the Central Yukon Field Office’s jurisdiction, this survey is pertinent.

3.0 Requirements

A complete LIDAR survey of the areas included in the table below and the attached map is requested in a 0.5-meter Digital Elevation Model form. Approximate acreage for each of the 6 areas is included with a 5-acre buffer on each, however, it was created in GIS online and the map may have inconsistencies, and the actual acreages may be larger or smaller. Each claim that is still under BLM management should be surveyed. After contract award, a full list of claims can be requested from the project manager. As the acreage being surveyed exceeds 2,000 acres and half of the groups are not accessible by terrestrial vehicle except in the winter, fixed wing LIDAR may be the most effective and efficient way to complete this project. However, a UAS based survey is also acceptable so long as it meets the requirements and budget of the project.

If fixed wing survey is conducted, there are airstrips at the Village of Ruby and Galena, and a fuel truck is available in Galena. Additionally, there is a small airstrip on state land at Long Creek, maintenance status varies as miners in the area use it on an undetermined basis.

Should a UAS based survey be selected, site access is difficult. The sites are remote and mostly abandoned, and the only 2 are confirmed accessible by 4-wheel drive, high clearance pick-up truck or similar highway vehicle or by OHV from Ruby, AK, are Long Creek and Greenstone Creek. The trail from Long Creek to Trail Creek has not been brushed out for many years and the condition is unknown. Miners access mines on State of Alaska managed land fairly regularly throughout the summer above the Sulatna Crossing, however, as these lands are not Federal, agreements may be needed to travel on mining access trails. The Ruby-Poorman Road is maintained by these miners and gets narrow and overgrown further south towards the Sulatna River as there are fewer active mines that way and therefore that area gets less frequent use. The Sulatna River crossing has been unusable for many years, preventing terrestrial access to Poorman Townsite, Solomon Creek, and Flat Creek in the summer months. Winter access to all sites is possible via snowmachine from Ruby, AK.

All sites are accessible by rotary-wing aircraft, though brushing of a landing zone may be required at some sites.

Area/ Group	Number of Claims	Location	Total Acreage for Area*
Flat Creek	4	KM T16S R17E Sec 31	100
Solmon Creek	1.5	KM T16S R17E Sec 30	41
Poorman Townsite	32	KM T16S R17E Secs 7, 15, 17, 18, 20, 21, 22	710
Greenstone Creek	6	KM T14S R17E Secs 4, 5, 8	160
Long Creek	31	KM T12S R17E Secs 28, 29, 32, 33 KM T13S R17E Secs 3, 4, 5, 8, 9, 17	750
Trail Creek	8	KM T13S R18E Secs 3, 9, 10, 16	360

*Areas estimated using eGIS and include a 5-acre buffer on each to account for errors in estimation

The contractor must provide all personnel, equipment, tools, materials, supervision, and any other items necessary to perform removal services as defined in this SOW. Any permits required must be obtained by the contractor prior to commencing field work.

The BLM project manager must furnish no equipment. After contract award, a schedule must be provided to the BLM project manager prior to the post-award kickoff meeting. After contract award, a map of the overall area as well as each smaller area may be requested from the BLM project manager, and a site visit can be scheduled with the BLM project manager prior to acceptance of a bid.

4.0 Deliverables

The Proposal and Final Report may be submitted as a hard copy or a digital copy to the BLM project manager. If submitting a hard copy, it should be mailed or dropped off at 222 University Drive, Fairbanks, AK 99709. DEM and raw imagery must be submitted as a digital copy to the BLM.

Project Proposal

The proposal must fully address the SOW, include the chosen methods for completing the project, and must abide by the requirements laid out in Section 3.0. The proposal must include expected personnel, anticipated timeline, equipment, and technical methodology for the project. Changes must be communicated to the BLM project manager for approval as soon as possible and before they are implemented. The Project Proposal must be submitted to the BLM project manager no later than 30 days prior to anticipated start date.

Survey Output

The survey output must be a 0.5-meter Digital Elevation Model (DEM) and must be furnished to the BLM project manager no later than 60 days prior to the end of the Period of Performance to allow for comments and revisions as necessary.

Final Report

The Final Report must provide a detailed description of all actions taken in association with the SOW, and any concerns identified during the survey. It must document days worked, personnel onsite, and equipment utilized. It must include any daily forms including tailgate meetings and Health & Safety meetings; the spatial data collected. It must also include explanations for any deviations from the projected schedule as provided prior to the kick-off meeting. The Report must be provided electronically to the BLM project manager no later than 30 days prior to the end of the Period of Performance or no later than one month after the work is completed.